

📖 Lesson 7 — Decision Trees 🌳

Now we're moving to another very important classification algorithm.

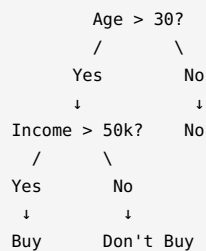
So far:

- Lesson 1 → ML Basics
- Lesson 2 → Linear Regression
- Lesson 3 → Regression Evaluation
- Lesson 4 → Classification
- Lesson 5 → Classification Evaluation
- Lesson 6 → KNN
- 📖 Lesson 7 → Decision Trees

1. What is a Decision Tree?

A Decision Tree makes predictions by asking a series of questions.

For example, suppose we want to decide whether a customer will buy a product:



The model essentially learns:

Which questions should I ask, and in what order, to separate the classes?

2. Real-Life Example 🌳

Imagine you're deciding whether to play cricket.

You might think:

```
Is it raining?
↓
Yes → Don't Play
No
↓
Is it windy?
↓
Yes → Maybe Don't Play
No
↓
Play
```

That's basically a decision tree.

```
Question
↓
```

```
Answer
↓
Another question
↓
Answer
↓
Final decision
```

3. Tree Structure

A Decision Tree contains several important parts.

Root Node

The first question:

Age > 30?

This is the root.

Branches

Possible answers:

```
Age > 30?
 /    \
Yes     No
```

Internal Node

Another question:

```
Age > 30?
 /
Yes
↓
Income > 50k?
```

Leaf Node 🍃

The final prediction:

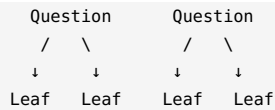
```
Income > 50k?
 /    \
Yes     No
↓       ↓
Buy     Don't Buy
```

The leaf contains the final class/prediction.

4. The Basic Structure

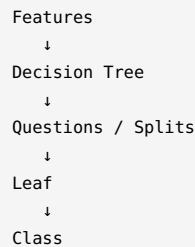
Remember this:

```
ROOT
↓
Question
 /  \
Yes  No
↓    ↓
```

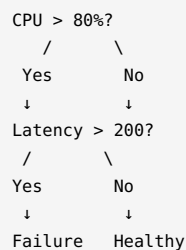


5. Decision Tree Classification

For classification:



Example:



This is particularly relevant to SentinelAI because your failure prediction problem can naturally be expressed through feature-based decisions.

6. Scikit-learn Decision Tree

Import:

```
from sklearn.tree import DecisionTreeClassifier
```

Create:

```
model = DecisionTreeClassifier()
```

Train:

```
model.fit(X_train, y_train)
```

Predict:

```
predictions = model.predict(X_test)
```

Evaluate:

```
from sklearn.metrics import accuracy_score

accuracy = accuracy_score(y_test, predictions)
```

```
print("Accuracy:", accuracy)
```

7. Complete Example

Let's use the same dataset we've been using.

```
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score

# Load dataset
data = load_breast_cancer()

X = data.data
y = data.target

# Split data
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

# Create model
model = DecisionTreeClassifier(random_state=42)

# Train
model.fit(X_train, y_train)

# Predict
predictions = model.predict(X_test)

# Evaluate
accuracy = accuracy_score(y_test, predictions)

print("Accuracy:", accuracy)
```

8. Do Decision Trees Need Scaling?

Here's something interesting.

KNN

```
Distance-based
↓
Scaling usually important
```

Decision Tree

```
Uses feature thresholds
↓
Scaling generally NOT required
```

For example, a tree might ask:

```
Age > 30?
```

or:

Salary > 50000?

It doesn't care that salary has larger numerical values in the same way KNN distance calculations do.

So:

Decision Trees generally don't require feature scaling.

9. How Does the Tree Decide What Question to Ask?

This is the most important part.

Suppose we have:

Age
Income
Experience

The tree needs to decide:

Which feature should I split on first?

For example:

Age > 30?

might separate the classes well.

But:

Experience > 5?

might separate them even better.

The algorithm evaluates possible splits and chooses one that makes the resulting groups more pure.

10. What Does "Pure" Mean?

Suppose a node contains:

[B] [B] [B] [B]

That's very pure.

All observations belong to the same class.

But:

[B] [R] [B] [R]

is mixed.

The tree wants to create groups that are as pure as possible.

```
Mixed data
↓
Split
↓
More pure groups
```

11. Gini Impurity

One common criterion used by Decision Trees is:

Gini impurity

Don't worry about the formula yet.

The important idea is:

```
Gini impurity
↓
Measures how mixed the classes are
```

Roughly:

```
Pure node
[B] [B] [B] [B]
↓
Low impurity

Mixed node
[B] [R] [B] [R]
↓
Higher impurity
```

Scikit-learn's default for `DecisionTreeClassifier` is typically:

```
criterion="gini"
```

So:

```
model = DecisionTreeClassifier(
    criterion="gini",
    random_state=42
)
```

12. Entropy

Another criterion is:

```
criterion="entropy"
```

You may also encounter:

```
criterion="log_loss"
```

For now, remember:

```
Decision Tree
↓
```

```
Need to choose best split
↓
Gini / Entropy / Log Loss
↓
Measure split quality
```

We'll go deeper into Gini and entropy later.

13. The BIG Problem With Decision Trees

Decision Trees can become too complicated.

Imagine:

```
Question
↓
Question
↓
Question
↓
Question
↓
Question
↓
Question
↓
Question
↓
Question
↓
...
```

The tree can memorize the training data.

This is called:

Overfitting

14. What is Overfitting?

Suppose:

```
Training accuracy = 100%
Testing accuracy  = 75%
```

The model has learned the training data extremely well...

but performs poorly on unseen data.

That's overfitting.

```
Training data
↓
Model memorizes
↓
Excellent training performance
↓
Poor unseen performance
```

15. Tree Depth

One of the most important controls is:

```
max_depth
```

For example:

```
model = DecisionTreeClassifier(  
    max_depth=3,  
    random_state=42  
)
```

This limits how deep the tree can grow.

Conceptually:

```
max_depth = 2
```

```
graph TD
    Root --> Node1
    Root --> Node2
    Node1 --> Leaf1
    Node1 --> Leaf2
    Node2 --> Leaf3
    Node2 --> Leaf4
```

A shallow tree is simpler.

16. Why Limit Tree Depth?

Without restrictions:

```
Tree becomes huge
    ↓
Can memorize training data
    ↓
Overfitting
```

With a reasonable depth:

```
Tree stays simpler
    ↓
Better generalization
```

This is called regularization or controlling model complexity.

17. Important Decision Tree Parameters

You'll frequently see:

max_depth

Controls maximum tree depth.

```
DecisionTreeClassifier(max_depth=5)
```

min_samples_split

Minimum number of samples required to split a node.

```
DecisionTreeClassifier(min_samples_split=10)
```

min_samples_leaf

Minimum samples allowed in a leaf.

```
DecisionTreeClassifier(min_samples_leaf=5)
```

criterion

How split quality is measured.

```
criterion="gini"
```

or:

```
criterion="entropy"
```

18. Visualizing a Decision Tree 🌲

Scikit-learn can actually show the tree.

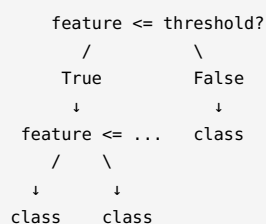
```
from sklearn.tree import plot_tree
import matplotlib.pyplot as plt

plt.figure(figsize=(20, 10))

plot_tree(
    model,
    filled=True
)

plt.show()
```

You'll see something conceptually like:



This is one reason Decision Trees are popular:

They can be relatively easy for humans to interpret.

19. A Very Important Difference

Compare KNN and Decision Tree.

KNN

```
New point
↓
Find nearest points
```

↓
Majority vote

Decision Tree

New point
↓
Question 1
↓
Question 2
↓
Question 3
↓
Leaf
↓
Prediction

20. Decision Tree Regression 🌲

Just like KNN, Decision Trees can also perform regression.

Classification:

```
from sklearn.tree import DecisionTreeClassifier
```

Regression:

```
from sklearn.tree import DecisionTreeRegressor
```

Classification

Tree
↓
Leaf
↓
Class

Regression

Tree
↓
Leaf
↓
Numerical value

For example:

House features
↓
Decision Tree
↓
₹52.4 lakh

21. Decision Tree vs KNN vs Logistic Regression

You now know three classification algorithms:

Algorithm	Main idea

Logistic Regression	Learn a mathematical decision relationship
KNN	Find nearby examples
Decision Tree	Ask learned questions/split data

Think of them like:

```

Logistic Regression
  ↓
Mathematical boundary

KNN
  ↓
Similarity / distance

Decision Tree
  ↓
Questions / rules

```

Lesson 7 — Must Remember

Decision Tree

A model that makes predictions by repeatedly splitting data using feature-based questions.

Root

First split/question.

Branch

Possible outcome of a split.

Leaf

Final prediction.

Gini / Entropy

Measures used to choose useful splits.

max_depth

Controls how deep the tree can grow.

Overfitting

A tree that's too complex can memorize training data.

Scaling

Decision Trees generally do NOT require feature scaling.

Exercises

Try these:

1. What is a Decision Tree?
2. What is the difference between a root node and a leaf node?

3. What does this mean?

```
DecisionTreeClassifier(max_depth=5)
```

4. Why can a very deep Decision Tree overfit?

5. Does a Decision Tree generally require `StandardScaler` ? Why?

6. What does Gini impurity measure?

7. What is the difference between:

```
DecisionTreeClassifier()
```

and

```
DecisionTreeRegressor()
```

8. Which algorithm uses distance?

```
KNN  
Decision Tree  
Logistic Regression
```